PAYTM BUILD FOR INDIA AI HACKATHON

Paytm ResolveOps

Autonomous FinTech Operations Teammate: Multi-Agent Dispute Resolution, Real-Time Banking Switch Reconciliation, and Indic Voice Execution.

TEAM SPOTPICX

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CORE STACK: LANGGRAPH • SARVAM AI • TOOL CALLING • FASTAPI • POSTGRESQL • DOCKER

The Friction Tax on India's Digital Payments

400% Peak Backlogs during High-Velocity Shopping

- Over 68% of support tickets originate from "Payment Debited but Not Credited" or delayed merchant QR settlements.
- During peak sales (Diwali, IPL), banking timeouts create up to 48–72 hour support backlogs, triggering user panic and chargebacks.

The Manual 5-Screen Swivel

- Human support executives spend 4–8 minutes per dispute manually verifying across 4–5 systems: NPCI switch logs, partner bank APIs, core ledgers, and fraud risk engines.
- Extremely repetitive, high-turnover L1 operational burden costing ₹25–₹35 per ticket.

Why Existing Chatbots Fail

- Traditional bots only recite passive FAQ scripts ("Please wait 24-48 business hours").
- They have zero agency: cannot query core switches, verify reversal eligibility, or trigger settlements.
- Non-English kirana merchants cannot effectively use complex text-based forms.

Paytm ResolveOps – An AI Teammate That Takes Action

Autonomous Agency, Not Just Chat

An intelligent teammate that holds actual operational authority to inspect internal switches, identify failure codes (e.g., U69 timeout, RB-01 declined), and execute ledger actions.

Closed-Loop Resolution in < 15 Seconds

- Reconciles transaction status in real-time across 3 nodes: Customer Bank $\leftrightarrow$ NPCI Switch $\leftrightarrow$ Paytm Merchant Ledger.
- Safely triggers reversal APIs or merchant balance adjustments without human intervention.

Native Indic Voice Copilot via Sarvam AI

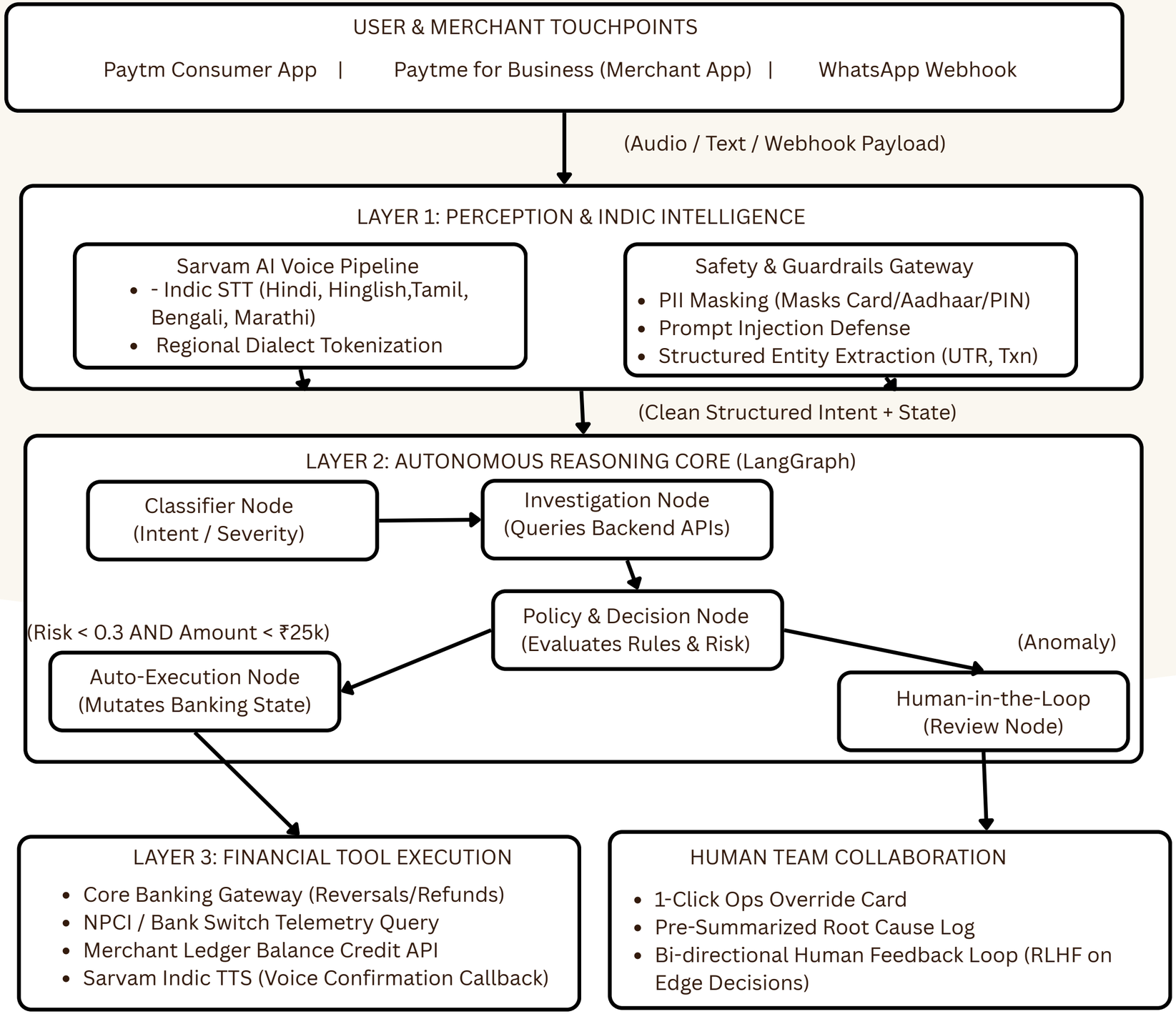
- Ingests native audio queries in Hindi, Tamil, Telugu, and Hinglish.
- Reassures kirana store merchants with clear voice updates directly over phone/Soundbox: "Sharma ji, aapka ₹1,450 HDFC server timeout ki wajah se atka tha. ResolveOps ne auto-refund initiate kar diya hai."

Collaborative Human-in-the-Loop (HITL) Safety Desk

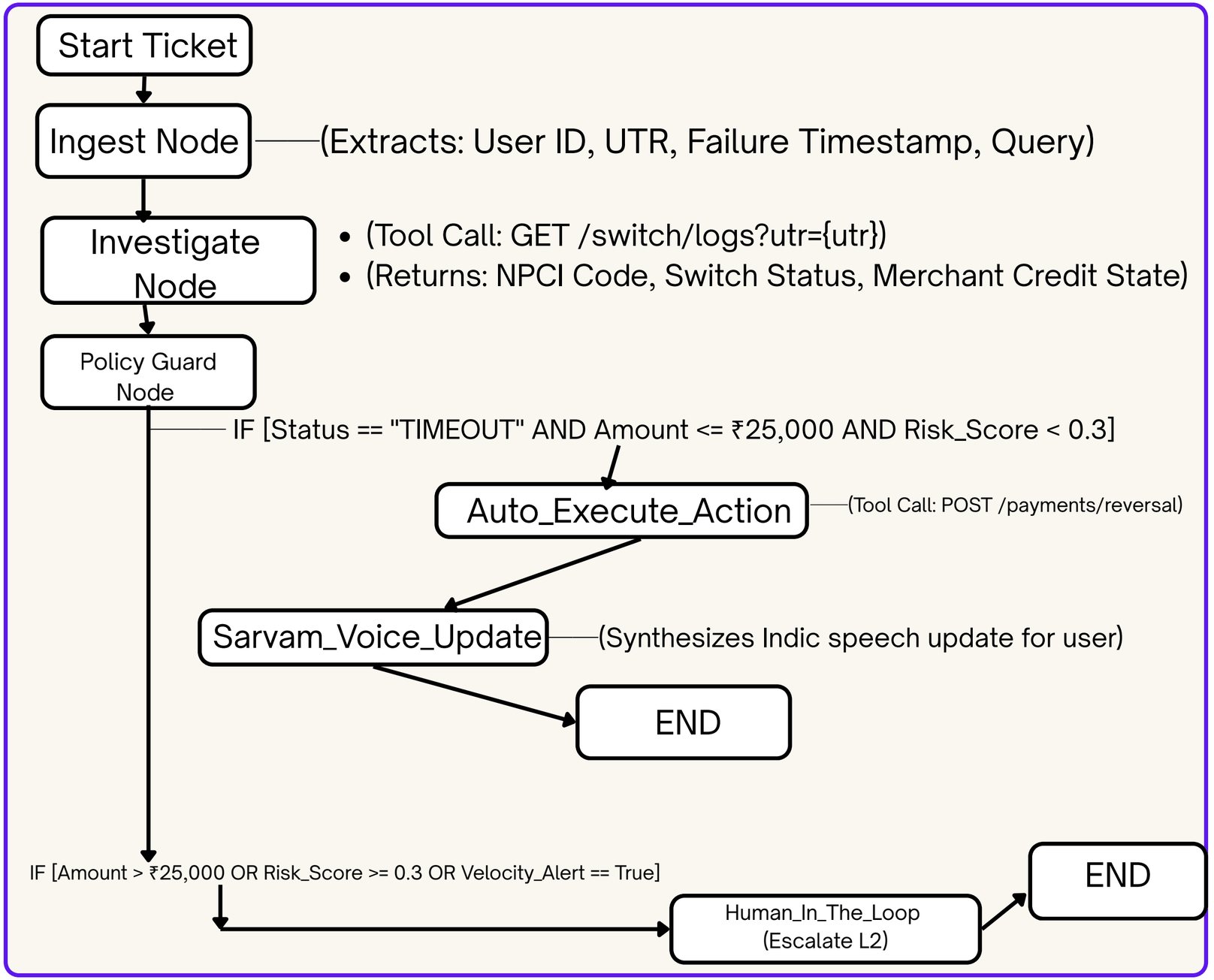
Flags high-value transactions (>₹25,000) or suspicious fraud patterns to human operators with pre-compiled root-cause dossiers for 1-click execution.



High-Level System Architecture



LangGraph Node State Transition Graph



USP (Unique Selling Proposition)

Action-First Agency vs. Conversational Chatbots:

Unlike LLM wrappers that merely respond with text, ResolveOps reads system logs, determines ground truth, and writes state back into core databases.

Deterministic Financial Safety:

Strict policy boundaries: LLMs reason and explain, but deterministic code controls all financial logic, limits, and authentication.

Tailored for Paytm's Merchant Network:

Directly solves the Soundbox & Kirana merchant communication barrier through Sarvam AI's Indic voice synthesis.

Seamless Human-AI Teammate Collaboration:

Bridges the gap between automation and human oversight: acts as an autonomous co-worker that does 80% of the manual digging and lets humans verify the rest.

Impact & Benefits

- **80% Ticket Deflection: Over 80% of routine payment failure disputes resolved end-to-end in under 15 seconds.**
- **70% Support Cost Reduction: Cuts L1 support agent seat licensing and BPO operational expenses by ₹18–₹25 per resolved dispute.**
- **Infinite Elastic Scalability: Handles 10x traffic bursts during festive seasons without hiring temporary customer care staff.**
- **Customer Retention & Trust: Replaces ambiguous 48-hour waiting times with immediate, transparent root-cause clarity in regional languages, mitigating user churn and bank chargebacks.**

Business Model & Scalability

Direct Internal Paytm ROI:

- Resolving 10M annual payment disputes autonomously translates to ₹150M+ in operational savings for Paytm.
- Prevents merchant churn to rival QR systems caused by delayed settlement anxieties.

B2B Fintech-as-a-Service (Paytm OpsAI Cloud):

- License the ResolveOps autonomous teammate engine to partner NBFCs, co-branded credit card issuers, and regional cooperative banks.
- Monetization: Per-Resolution SaaS fee (₹2.50 to ₹4.00 per auto-resolved ticket vs. ₹30+ for human outsourced BPO).

Future Scalability Roadmap:

- Phase 1 : Payment Recon & UPI Dispute Settlement.
- Phase 2: Autonomous Merchant Loan Document verification & collection follow-ups.
- Phase 3: Soundbox-integrated voice dispute alerts directly in merchant retail shops.